Campus Safety & Incident Analytics System

Entity-Relationship (ER) Design Report

Alex Ross, Jasmine Alabi, Aastha Paranthaman, Gayatri Kamtala

Point of contact email addresses: walexross2022@vt.edu, ajasmine@vt.edu, aasthap@vt.edu, gayatrikam@vt.edu

CS 4604, Dr. Nizamani

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1. Problem Definition

This system provides a centralized platform to track and analyze campus crime data for Virginia Tech. Currently, crime data is published in PDF reports that require users to manually read through tables and interpret information on their own. This process is time-consuming, inconsistent, and makes it difficult to identify patterns or trends in campus safety.

To address this issue, the system organizes campus crime data into a structured relational database. Incident data is extracted from official campus police PDF reports using a custom Python-based parser and then inserted into the database. By standardizing the format of incident data, the system enables efficient querying, filtering, and analysis. Users can easily view incidents, explore trends over time, and access summarized insights through a web-based interface.

Campus crime reports often contain similar information but are presented in slightly different formats. This system extracts key fields such as case number, date reported, occurrence date and time, offense type, location, and disposition, and stores them in a consistent schema. This standardization improves data quality and allows for meaningful comparisons and analysis within the dataset.

The system focuses on post-COVID campus crime data, when campuses resumed normal operations, providing relevant and recent insights into campus safety. The structured database supports filtering by offense type, date, and location, enabling users to better understand patterns in reported incidents.

While the current implementation focuses on a single campus, the database design is structured in a way that could support expansion to additional campuses in the future with minimal changes.

2. System Users and Privileges

User Type 1: Regular User (Students, Parents)

Privileges:

- Create an account and log into the system
- View publicly available incident records
- Filter and analyze data by offense, location, and date
- View charts and summary statistics

User Type 2: Administrator

Privileges:

- Full access to system functionality
- Add, edit, and delete incident records
- Manage and maintain database integrity
- Trigger or manage data ingestion processes

User Type 3: Data Analyst

Privileges:

- Perform advanced analysis on incident data
- Validate data completeness and consistency
- Generate reports and summaries

3. Entities and Their Descriptions

Entity 1: Incident

Description:

Represents a single reported campus crime case documented by the Virginia Tech Police Department. Each incident corresponds to a unique case number and captures when the crime was reported, when it occurred, where it occurred, and its current disposition. This entity serves as the central record for all crime-related information in the system.

Entity 2: Offense

Description:

Represents a specific type of criminal offense, such as underage possession of alcohol, larceny, assault, or burglary. This entity standardizes offense classifications and allows multiple incidents to reference the same offense type, supporting consistent reporting and analysis.

Entity 3: IncidentOffense

Description:

Represents the association between incidents and offenses. Since a single incident may involve multiple offenses, this entity resolves the many-to-many relationship between Incident and Offense, ensuring that all offenses related to a case are accurately recorded.

Entity 4: Location

Description:

Represents the physical location where an incident occurred, including campus buildings, parking lots, streets, intersections, and outdoor areas. This entity allows incidents to be spatially organized and analyzed based on where crimes occur on or near campus.

Entity 5: Disposition

Description:

Represents the status or outcome of an incident, such as active, inactive, cleared by arrest, or referred to student conduct. This entity standardizes case outcomes and enables consistent tracking of how incidents are resolved over time.

Entity 6: ReportingAgency

Description:

Represents the organization responsible for reporting or handling an incident. In this system, it primarily refers to the Virginia Tech Police Department but is designed to support additional agencies if needed in the future.

Entity 7: User

Description:

Represents authenticated users of the system. Users can create accounts, log in, and interact with the system based on their assigned role. The system supports role-based access control, where administrators have permissions to modify database records, while regular users are restricted to viewing and analyzing incident data.

4. Relationships and Their Descriptions

Relationship 1: Incident - Occurs At - Location

This relationship represents the physical place where a reported crime occurred. It exists because every incident must happen at a specific location, such as a parking lot, street, campus building, or outdoor area, and enables spatial organization of crime data. By linking incidents with locations, the system can support crime mapping and location-based statistical reporting.

Relationship 2: Incident - Has - Disposition

This relationship represents the official status or resolution of a crime case. It exists because each incident progresses through an investigative process and is assigned a standardized outcome such as active, inactive, cleared by arrest, and referred to student conduct. This relationship ensures uniform reporting, consistent tracking of case resolution, and allows analysis of how incidents are resolved over time.

Relationship 3: Incident - Reported By - ReportingAgency

This relationship represents the organization that is responsible for handling or documenting a reported crime. This relationship is needed to show that each incident is associated with a responsible reporting authority within the system.

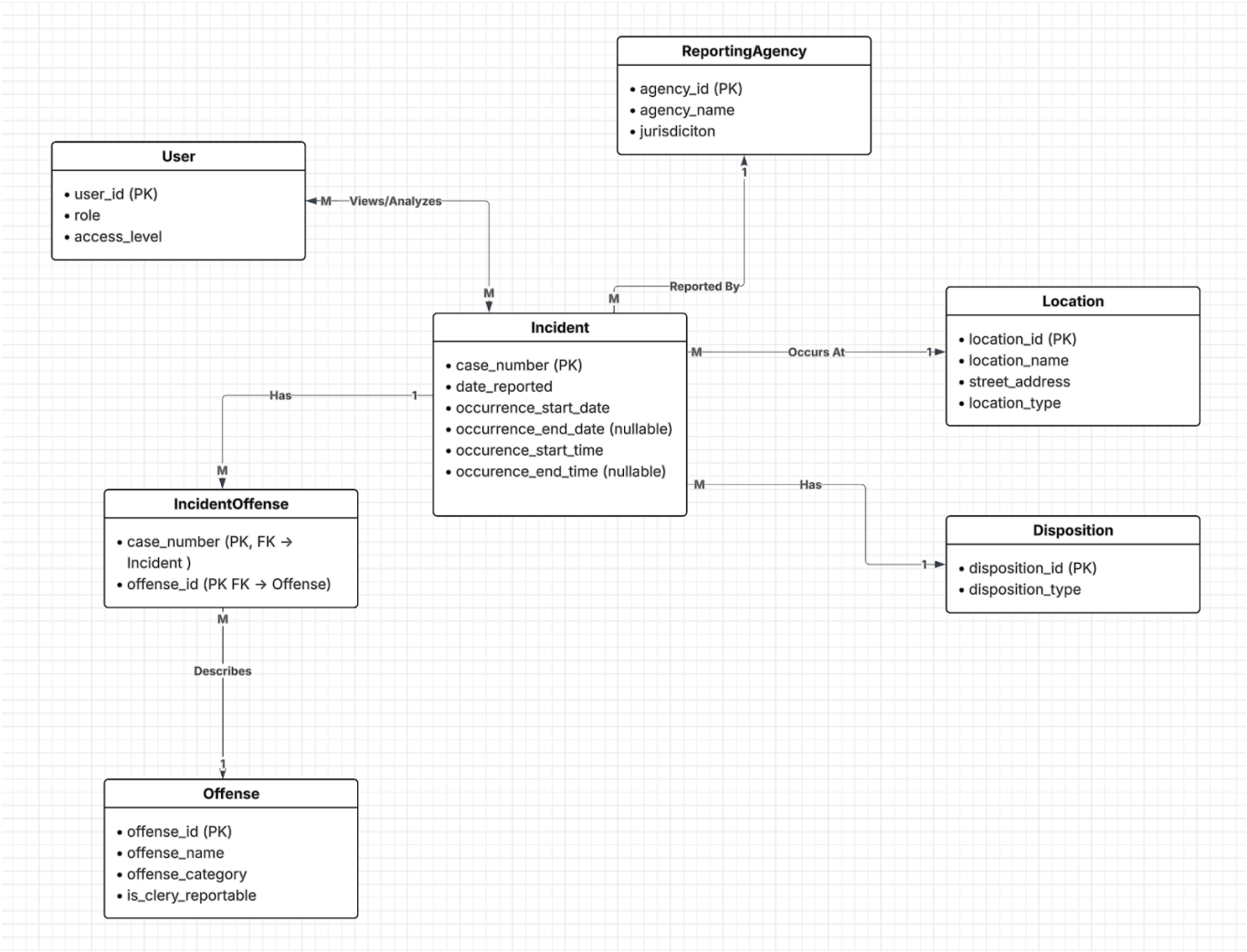
Relationship 4: Incident - Has - IncidentOffense

This relationship represents the association between an incident and the offenses involved in that incident. It exists because a single incident may involve one or multiple types of criminal offenses. This relationship allows each incident to be linked to all applicable offense records and supports accurate crime classification and reporting.

Relationship 5: IncidentOffense - Describes - Offense

This relationship resolves the many-to-many relationship between Incident and Offense. It exists because an incident may involve multiple offenses, and a single offense type may appear in many incidents. This relationship ensures accurate offense classification, prevents duplication of offense records, and maintains data integrity.

Relationship 6: User - Interacts With -Incident

This relationship represents authenticated interaction between users and the system. Users can log in to access incident data, perform analysis, and, depending on their role, modify records. This relationship is enforced at the application level through role-based access control implemented in the Flask backend.

5. Entity Relationship Diagram (ERD)